

A 28GHz Quadrature Up-conversion Transmitter in 65nm CMOS for 5G mmWave Radio

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Abstract—A 28GHz RF transmitter is designed in 65nm CMOS for 5G millimeter-wave radio applications. The transmitter is based on a direct-conversion quadrature up-conversion architecture for achieving small silicon area and low power dissipation. The image suppression is improved by adopting I/Q LO mismatch calibration at the LO driving buffer of mixer. The power amplifier is designed in a differential type with cross-coupled neutralized capacitors for stability and reverse-isolation improvement. Combined schematic and EM simulation results show that the transmitter gives the power gain of 37 dB, OP1dB of +5.8 dBm, and P_{sat} of +11.6 dBm. The total transmitter consumes 111 mW from a 1.2 V supply and occupies 1.78 mm².

Keywords: RF transmitter, quadrature up-conversion, power amplifier, 5G, 28GHz, CMOS

I. INTRODUCTION

Envisioning the enhanced mobile broadband over the air in mobile condition, use of millimeter-wave frequency bands above 6GHz becomes a key enabling technology in the fifth generation (5G) cellular communication. The U.S. and Korea plans to deploy the mm-wave radio at 28GHz band of 26.5-29.5GHz. Various research and development activities are currently going on worldwide for this. Compared to the path loss in the legacy 2.5GHz band defined by the 3GPP (3rd Generation Partnership Project), the signal path loss in the 28GHz band is 20-25 dB higher. In addition, an additional loss of about 5 dB is added by shadowing effect of electromagnetic waves being absorbed through the human hand, body, building wall, etc. Such large attenuation in the mm-wave band requires higher gain and output power in the RF transmitter circuit.

Mm-Wave RF transmitter can be based on either two-step heterodyne conversion [1] or direct conversion architecture [2]. Two-step conversion architecture can be advantageous in terms of possibly reusing and easy interfacing with legacy RF/IF transceivers, but high circuit complexity and power consumption are downside. On the other hand, the direct conversion architecture is advantageous for simpler circuits, smaller silicon area, lower power consumption, lower noise, higher linearity, and no need of additional RF/IF transceivers but a direct interface with a digital baseband.

We present a CMOS design of a low-power 28-GHz quadrature direct up-conversion transmitter circuit for mm-wave 5G radio applications.

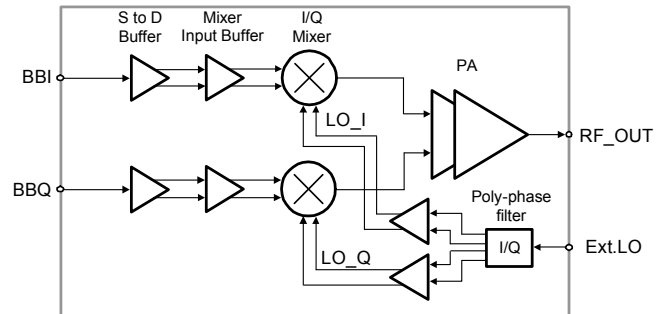


Fig. 1 Block diagram of 28 GHz direct-conversion transmitter.

II. CIRCUIT DESIGN AND RESULTS

Fig. 1 shows the block diagram of the RF transmitter of this work. In the quadrature up-conversion architecture, the I/Q mismatch can result in severe image and LO leakage components, which deteriorates the EVM performance of the output signal. In this design, I/Q mismatch calibration block is implemented at the mixer's I/Q LO buffer stage. Thus, the image leakage can be properly suppressed.

For the quadrature LO generation, an external single-ended LO signal is fed to a transformer baluns to be converted to differential. Subsequently a RC polyphase filter transforms it to I/Q differential signal, and finally fed to the I/Q LO buffer stage. The LO buffer has about 10 dB gain to compensate for the polyphase filter induced loss. The single-ended baseband signal with its frequency up to 1 GHz is fed in I/Q form, and converted to differential by using the single-to-differential buffer. The up-converted signal at 28 GHz is fed the power amplifier to deliver a nominal output power of 0 dBm.

Fig. 2 shows the schematic of the I/Q up-conversion mixer. The mixer is based on a gilbert-cell architecture for small LO leakage and high gain [3]. The LO signal amplitude driven by the LO buffer is about 200 mV_{pp} (-10 dBm), which is sufficient to drive the switching stage of the mixer. The inductive load L_{1-2} of 210 pH is used for guaranteeing tuned characteristic at around 28 GHz at the mixer output.

Fig. 3 is the schematic of the power amplifier. The power amplifier dominates the total linearity and efficiency in the RF transmitter, thus needs the most special attention in the design. The common-source type is employed rather than a cascode because it benefits to low voltage operation [4]. However, the common-source type can be vulnerable to stability issue due to

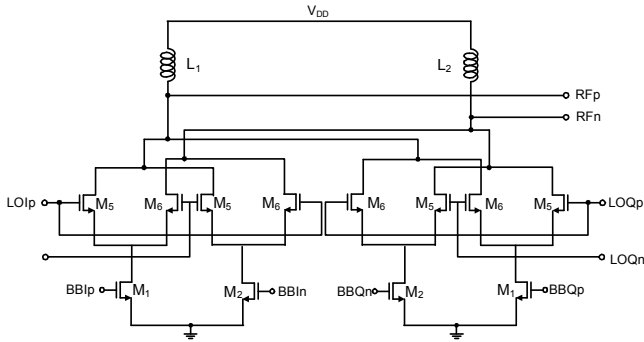


Fig. 2 Image-rejection mixer schematic

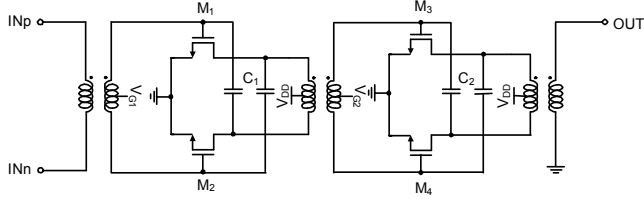


Fig. 3 Power amplifier schematic

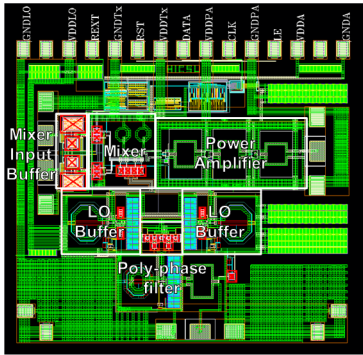


Fig. 4 Top layout

the unwanted feedback through the gate-drain capacitance of FET. This issue is alleviated by neutralize or compensate the feedback effect through the neutralized cross-coupled capacitors. The neutralized capacitor provides compensating current to cancel the unwanted feedback effect through C_{gd} . The neutralized capacitors C_{1-2} are employed both in the first and second stages, and is optimized to maximize the reverse isolation of the core stage for improved stability [5]. Accordingly, the neutralized MOM capacitors for the first (C_1) and second (C_2) stages are set to 15 fF and 19 fF, respectively. The gate width of M_{1-2} of the first stage is 50 μm (10 x 5 μm), and the gate width of M_{3-4} of the second stage is 80 μm (16 x 5 μm). The power amplifier employs three transformer baluns. The transformer baluns are designed in a rectangular stacked structure. Compared to the conventional interleaving structures, it has higher coupling factor and self-resonance frequency, and occupies less area. The three transformer baluns are independently optimized through EM simulations for optimum gain, output power, and impedance matching performances.

The transmitter is designed and laid out in 65nm CMOS 1P9M process. Fig. 4 shows the top layout view of the total circuit. The active area is 1450 x 1240 μm^2 . The routings of LO and RF signal lines are carefully placed for maximum symmetry and minimum mismatch. All inactive empty spaces are filled

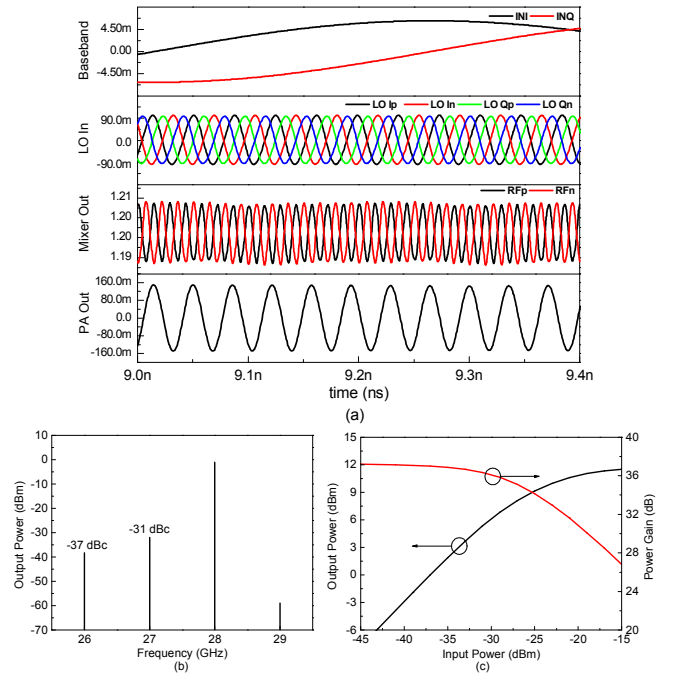


Fig. 5 (a) Transient simulation result. (b) Output spectrum. (c) Output power & power gain against input power.

with top metal to minimize parasitic impedances imposed on V_{DD} and ground rails. Fig. 5 depicts the simulation results of the combined schematic and EM simulations. Fig. 5(a) shows the time-domain waveforms of the baseband I/Q input signal (9 mV_{pp}), differential I/Q LO signals (200 mV_{pp}), the mixer output signal (20 mV_{pp}), and finally the power amplifier output signal (300 mV_{pp}). The total voltage gain is +30.4 dB. Fig. 5(b) shows the output spectrum at the power amplifier output. The LO and image leakage components are -31 dBc and -37 dBc, respectively, while the output power is 0 dBm. Fig. 5(c) shows the overall power gain of 37 dB, OP1dB of +5.8 dBm, and P_{sat} of +11.5 dBm. The total power consumption of the transmitter is 111 mW and the supply voltage is 1.2 V.

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